PROJECT DOCUMENTATION

Project title

MicroGoal – Smart Goal Management System

1. Introduction

Team ID: SWTID-2026-7996

Team Leader :
NIRMALA G (nirmalagvan@gmail.com)

Team Members:
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Roles and Responsibilities

Team Leader – Nirmala G

- Oversees backend architecture and database design.
- Manages API structure and integration.
- Ensures project milestones are completed on time.

Backend Developer – Piyadharshini M

- Develops RESTful APIs using Node.js and Express.js.
- Implements authentication and authorization.
- Handles business logic and server-side validation.

Database Manager – Kowsalya N

- Designs MongoDB schema and models.
- Ensures data integrity and optimization.
- Manages indexing and performance tuning.

Testing & Documentation Lead – Swetha A

- Performs API testing using Postman.
- Handles debugging and error handling.
- Prepares technical documentation.

2. Project Overview

MicroGoal – Smart Goal Management System is a backend-focused application designed to help users create, track, and manage personal or professional goals efficiently.

The system provides secure user authentication, goal tracking features, progress monitoring, and deadline management using a scalable backend architecture built with Node.js and MongoDB.

Main Goals of the Project :

1. **Secure User Management** – Allow users to register, login, and manage their profiles securely.
2. **Goal Creation & Tracking** – Enable users to create short-term and long-term goals with deadlines and progress status.
3. **Progress Monitoring** – Track completion percentage and update goal status dynamically.
4. **Scalable Backend Architecture** – Ensure high performance using Node.js and MongoDB.
5. **Data Security & Integrity** – Implement JWT authentication and secure database storage.

3. Architecture

MicroGoal follows a **three-tier architecture** for better scalability and maintainability.

1. Presentation Layer (Client / Frontend)

- Sends HTTP requests to backend APIs.
- Displays user goals, progress, and profile details.

2. Application Layer (Backend – Node.js / Express.js)

- Handles business logic.
- Processes API requests and responses.
- Manages authentication and validation.

3. Data Layer (MongoDB Database)

- Stores user accounts and goals.
- Maintains structured collections.
- Provides fast querying and indexing.

4. Setup Instructions

Prerequisites

Before setting up MicroGoal Backend, install:

- Node.js
- MongoDB
- Git
- Postman
- Visual Studio Code

Installation Steps

1. Clone the Repository:

```
git clone <repository-url>
```

2. Navigate to Project Folder:

```
cd microgoal
```

3. Install Dependencies:

```
npm install
```

4. Create .env File and Add:

```
PORT=5000
MONGO_URI=your_mongodb_connection_string
JWT_SECRET=your_secret_key
```

5. Start the Server:

```
npm start
```

6. Server Runs On:

http://localhost:5000

5. Folder Structure

```
microgoal/
│── server.js
│── config/
│   └── db.js
│── models/
│   ├── User.js
│   └── Goal.js
│── routes/
│   ├── userRoutes.js
│   └── goalRoutes.js
│── controllers/
│   ├── userController.js
│   └── goalController.js
│── middleware/
│   └── authMiddleware.js
│── .env
│── package.json
```

Explanation

- **models/** → MongoDB schemas using Mongoose
- **routes/** → API endpoints
- **controllers/** → Business logic
- **middleware/** → Authentication and validation
- **config/** → Database connection setup

6. Running the Application

Start Backend Server

1. Navigate to project folder:

```
cd microgoal
```

2. Run server:

```
npm start
```

Or for development mode:

```
npm run dev
```

3. Open in browser or Postman:

http://localhost:5000

7. API Documentation

User APIs

Register User
POST /api/users/register

- Registers a new user (name, email, password).

Login User
POST /api/users/login

- Authenticates user and returns JWT token.

Get Profile
GET /api/users/profile

- Returns logged-in user details (Protected Route).

Goal APIs

Create Goal
POST /api/goals

- Create a new goal with title, description, deadline.

Get All Goals
GET /api/goals

- Retrieves all goals of logged-in user.

Update Goal
PUT /api/goals/:id

- Updates goal details or progress.

Delete Goal
DELETE /api/goals/:id

- Deletes a specific goal.

8. Authentication

MicroGoal uses **JWT (JSON Web Token)** authentication.

Secure Login

- User logs in with email and password.
- Server generates a JWT token.
- Token is sent in response.

Protected Routes

- Token must be included in request header:

  Authorization: Bearer <token>

- Middleware verifies token before allowing access.

9. User Interface

Since this project focuses on backend development, the system supports integration with any frontend (React, Angular, or Mobile App).

Main expected UI pages:

- Login Page
- Registration Page
- Dashboard (View Goals)
- Create Goal Page
- Profile Page

10. Testing

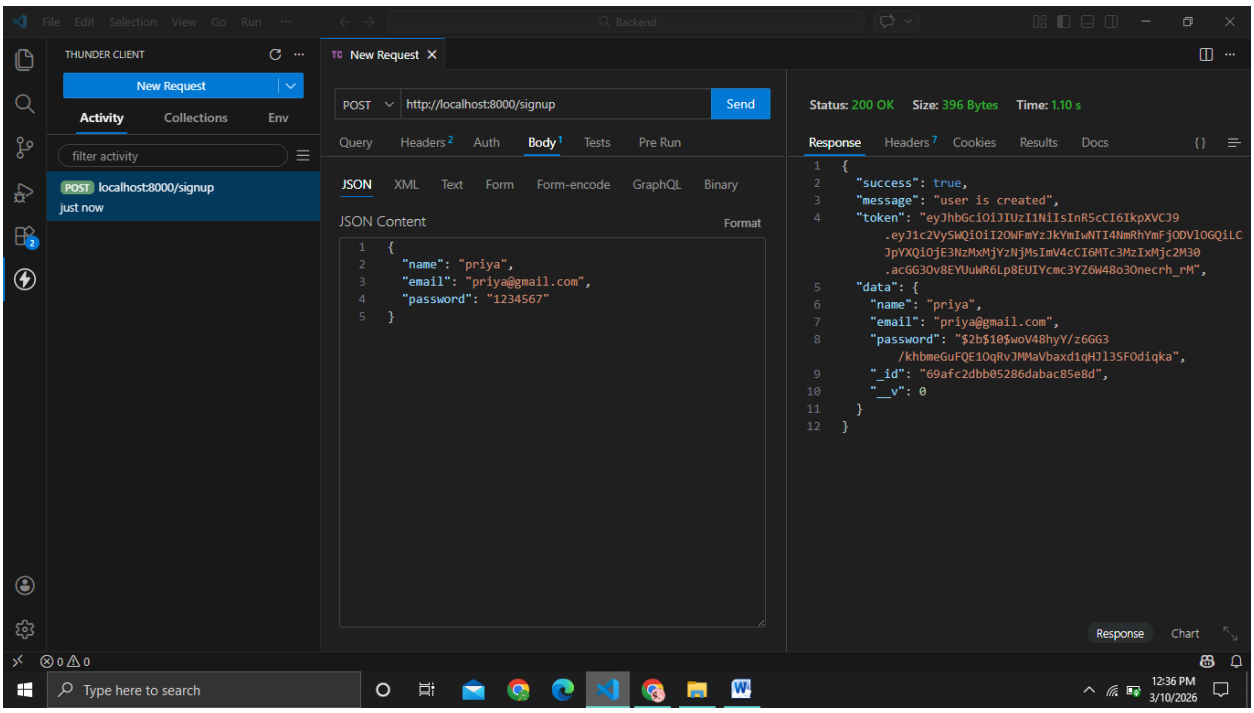
Manual Testing

- Tested each API endpoint using Postman.
- Verified authentication and protected routes.
- Checked CRUD operations for goals.

Testing Tools

- Postman (API Testing)
- MongoDB Compass (Database Inspection)
- Chrome Dev Tools (If frontend connected)

11. Screenshots



12. Known Issues

- No email verification system implemented.
- Limited password strength validation.
- No rate limiting for login attempts.
- Basic error handling messages.

13. Future Enhancements

1. Email Verification & Password Reset
2. Goal Categorization & Tags
3. Reminder Notifications
4. Analytics Dashboard for Progress Tracking
5. Role-Based Access Control (Admin/User)
6. Deployment on Cloud (AWS / Render / Railway)